The $\pi\Delta$ splitting function in PionCloud: comparison with McKenney *et al.* (2016)

PionCloud maintainers (TDIS) — technical note

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Abstract

For the πN channel, PionCloud reproduces the pion multiplicities of McKenney, Sato, Melnitchouk and Ji [1] to 3%. For the $\pi\Delta$ channel it gives $\langle n\rangle_{\pi\Delta} = 0.027$ where Ref. [1] quotes 0.06 for the same regulator and cut-off, a factor 2.2. We show that this is a difference of *model*, not a bug: implementing Eq. (5) of Ref. [1] with the same regulator reproduces its 0.06, and the factor splits into $\times 1.5$ from the isospin (charge-state) bookkeeping and $\times 1.43$ from the functional form of the splitting function (the "Sullivan" Δ-pole form of the original Fortran vs. the chiral-EFT on-shell form of Ref. [1]). The couplings agree to 4%. **Decision:** PionCloud keeps the Fortran (Hobbs/HSS) $\pi\Delta$ model, which is the one used for the TDIS proposal; this note documents the difference.

1 Context

PionCloud is a C++ port of the pion-cloud codes of T. Hobbs and W. Melnitchouk (`3Var_x.f`, `3Var_theta.f`, `f2_pi_sub.f90`) used for the TDIS proposal PR12-15-006 [4]. It reproduces the proposal's figures, including the $\pi\Delta$ curves of its Figs. 6 and 37, to about 1%. As an independent check of the normalisation of the splitting functions $f_{MB}(y, k_\perp)$ (per $dk_\perp^2$, with y the meson light-cone fraction), we computed the average multiplicities

$$\langle n\rangle_{MB} = \int_0^1 dy \int_0^\infty dk_\perp^2 \, f_{MB}(y, k_\perp), \qquad \text{summed over charge states,} \tag{1}$$

and compared them with Table I of Ref. [1], which lists them for several regulators fitted to the E866 $\bar{d} - \bar{u}$ data. All numbers below use the s-dependent exponential regulator $F = \exp[(M^2 - s)/\Lambda^2]$ with $s = (k_\perp^2 + m_\pi^2)/y + (k_\perp^2 + M_B^2)/(1-y)$, which is Eq. (12) of Ref. [1] and `form_factor = exp_s` in PionCloud (the same convention).

2 Results

	PionCloud	Ref. [1], Table I	ratio
$\langle n\rangle_{\pi N}$, `exp_s`, $\Lambda = 1.33$ GeV	0.224	0.23	0.97
$\langle n\rangle_{\pi N}$, Pauli–Villars, $\Lambda = 0.27$ GeV	0.310	0.31	1.00
$\langle n\rangle_{\pi\Delta}$, `exp_s`, $\Lambda = 1.33$ GeV	0.027	0.06	0.45

The πN agreement validates the πN splitting function, its normalisation, the isospin sum (charge exchange $\times 2$ + neutral $\times 1$) and the Pauli–Villars implementation (Eq. (9) of Ref. [1]). The $\pi\Delta$ value differs by a factor 2.2.

3 The two $\pi\Delta$ splitting functions

PionCloud / original Fortran (`fypiD` in `include/SplittingFunctions.hh`, unchanged from the Fortran). With $\bar{y} \equiv 1 - y$ the baryon fraction,

$$f^{\mathrm{F}}_{\pi\Delta}(y, k_\perp) = \frac{2}{3}\,\frac{g^2}{16\pi^2 m_\pi^2}\,\frac{y}{\bar{y}}\,\frac{\left[k_\perp^2 + (M_\Delta - \bar{y}M)^2\right]\left[k_\perp^2 + (M_\Delta + \bar{y}M)^2\right]^2}{6M_\Delta^2\,\bar{y}^3\,y^2\,(s - M^2)^2}\,F^2, \qquad \frac{g^2}{4\pi} = 0.2237. \quad (2)$$

This is the standard time-ordered ("Sullivan") form evaluated at the Δ pole, as in Holtmann, Szczurek and Speth [2] and Melnitchouk, Speth and Thomas [3]: at large $k_\perp$ the numerator grows like $k_\perp^6$. The prefactor $\frac{2}{3}$ is applied in the Fortran for `dis = 0` ("charge exchange"); for `dis = 1` the coupling is set to zero. The charge-state content of the factor $\frac{2}{3}$ is not documented in the Fortran.

McKenney *et al.* [1], Eqs. (4)–(7). The on-shell part of the $p \to \pi^+\Delta^0$ splitting function in chiral effective theory,

$$f^{(\mathrm{on})}_\Delta(y) = C_\Delta \int dk_\perp^2\,\frac{y\,(\bar{M}^2 - m_\pi^2)}{(1-y)\,D_{\pi\Delta}^2}\Big[(\bar{M}^2 - m_\pi^2)(\Delta^2 - m_\pi^2) - \big(3(\Delta^2 - m_\pi^2) + 4MM_\Delta\big)D_{\pi\Delta}\Big]F^2, \quad (3)$$

with $\bar{M} = M_\Delta + M$, $\Delta = M_\Delta - M$, $D_{\pi\Delta} = -[k_\perp^2 - y(1-y)M^2 + yM_\Delta^2 + (1-y)m_\pi^2]/(1-y)$, $C_\Delta = g^2_{\pi N\Delta}/[(4\pi)^2\,18M_\Delta^2]$ and $g_{\pi N\Delta} = (3\sqrt{2}/5)\,g_A/f_\pi \approx 11.8$ GeV^{-1}. The charge states are related by $2f_{\pi^-\Delta^{++}} = 3f_{\pi^0\Delta^+} = 6f_{\pi^+\Delta^0}$, and $\langle n\rangle_{\pi\Delta} = 6\int dy\, f^{(\mathrm{on})}_\Delta$. The authors note explicitly that Eq. (3) "differs from the 'Sullivan' form often used in the literature", which has a higher power of $k_\perp$ and neglects end-point contributions. Their δ-function and end-point terms (at $y = 0$ and $y = 1$) do not enter $\langle n\rangle$ as defined in their Eq. (7).

4 Decomposition of the factor 2.2

We implemented Eq. (3) with the same regulator (program `docs/notes/piDelta_check.cc`). At $\Lambda = 1.33$ GeV:

	$\langle n\rangle_{\pi\Delta}$	step
PionCloud = Fortran, Eq. (2) with the factor $\frac{2}{3}$	0.027	
same functional form, all charge states with total weight 1 ($\times\frac{3}{2}$)	0.040	$\times 1.50$
Eq. (3) $\times 6$, $g_{\pi N\Delta} = 11.56$ GeV^{-1} (from g_A, f_π)	0.058	$\times 1.43$
Eq. (3) $\times 6$, $g_{\pi N\Delta} = 11.8$ GeV^{-1} (as quoted)	0.060	
Ref. [1], Table I	0.06	

- Our implementation of Eq. (3) reproduces the published 0.06, so the comparison is under control.
- **Couplings** are equivalent: $g^2/4\pi = 0.2237/m_\pi^2 = 11.5$ GeV^{-2} (Fortran) vs. $11.8^2/4\pi = 11.1$ GeV^{-2} (Ref. [1]), a 4% difference.
- **Isospin bookkeeping** ($\times 1.5$): if the Fortran factor $\frac{2}{3}$ is read as a partial charge-state sum relative to a total weight of 1, the total is $\frac{3}{2}$ of the PionCloud value. This step is an interpretation, since the meaning of $\frac{2}{3}$ is not documented.
- **Functional form** ($\times 1.43$): the chiral-EFT on-shell form gives more strength than the Sullivan form at the same Λ; it is also slightly softer: $\langle y\rangle = 0.286$, $\langle k_\perp^2\rangle = 0.117$ GeV^2 vs. 0.316 and 0.136 GeV^2 for Eq. (2).
- At the HSS value $\Lambda_{\pi\Delta} = 1.39$ GeV (the PionCloud default): 0.034 (PionCloud), 0.050 ($\times\frac{3}{2}$), 0.071 (Eq. 3).

5 Conclusion and recommendation

The difference is a difference between two models of the $\pi\Delta$ fluctuation, not an error in PionCloud. The cut-off values of Ref. [1] were fitted *with their* functional form and should not be combined with Eq. (2); conversely, the HSS values ($\Lambda_{\pi\Delta} = 1.39 \pm 0.07$ GeV) belong to the Sullivan form used in the TDIS proposal. PionCloud keeps the Fortran model. A user who wants the chiral-EFT form can implement Eq. (3) as a `SplittingModel` (`include/Models.hh`); `piDelta_check.cc` contains a ready implementation of the integrand. Open point: the exact charge-state meaning of the Fortran factor $\frac{2}{3}$ (relevant for the absolute $\pi\Delta$ normalisation in the tagged structure functions).

Reproducing the numbers.

```
g++ -std=c++17 -O2 -Iinclude docs/notes/piDelta_check.cc -o piDelta_check
./piDelta_check        # header-only, no CTEQ6/ROOT needed, ~1 min
```

References

[1] J. R. McKenney, N. Sato, W. Melnitchouk and C.-R. Ji, Phys. Rev. D **93**, 054011 (2016), arXiv:1512.04459.

[2] H. Holtmann, A. Szczurek and J. Speth, Nucl. Phys. A **596**, 631 (1996).

[3] W. Melnitchouk, J. Speth and A. W. Thomas, Phys. Rev. D **59**, 014033 (1999).

[4] A. Camsonne *et al.*, JLab proposal PR12-15-006 (2015), approved as E12-15-006.

[5] T. J. Hobbs, Few-Body Syst. **56**, 363 (2015), arXiv:1411.2216.